

ATHARV PAWAR

SOFTWARE & AI ENGINEER

Email: pawaratharv06@gmail.com | Phone: +91 8010562215 | Location: India (Open to Remote / Relocation)

Portfolio: atharvpawar.work.gd | GitHub: github.com/Atharv06pawar | LinkedIn: linkedin.com/in/atharvpawar

PROFESSIONAL SUMMARY

End-to-End **Software & AI Engineer** and technical generalist with hands-on experience taking complex problems from concept and system architecture to backend implementation, AI/ML integration, testing, and deployment. Proven expertise across LLM systems, grounded scientific RAG pipelines, custom PyTorch ML models, modular REST APIs, real-time WebSocket audio streaming, and responsive web platforms. Founder of REDAESTH with demonstrated ownership in solving ambiguous technical problems and delivering reliable, production-ready software.

TECHNICAL SKILLS & DOMAIN EXPERTISE

Languages:	Python, JavaScript (ES6+), TypeScript, C++ (C++17), Java, HTML5, CSS3, SQL
AI & Machine Learning:	LLM Systems, RAG Pipelines, FAISS (C++ Vector DB), Dense Embeddings (SentenceTransformers), PyTorch (Custom Transformers, RoPE, SwiGLU, RMSNorm, KV Cache), Dataset Engineering, Multi-Agent Systems, AI Red Teaming
Backend & APIs:	Node.js, Express, RESTful APIs, JWT Authentication, Microservices Architecture, Serverless (Vercel), Middleware
Real-Time & Voice:	WebSockets, Bidirectional Audio Streaming, Voice Activity Detection (VAD), STT/TTS Integration
Frontend & UI:	React, Next.js, Tailwind CSS, Responsive Layout Architecture, State Management, DOM APIs, Canvas
Data & Storage:	PostgreSQL, MongoDB, SQLite (Android Room Database), Vector Stores (FAISS), Persistent JSON Context Stores
DevOps & Security:	Git/GitHub, Docker Containerization, Linux Bash Scripting, CI/CD Workflows, Vercel, API Security, Prompt Injection Defense

PROFESSIONAL & FOUNDER EXPERIENCE

Founder & Lead Systems Engineer REDAESTH Pvt. Ltd.	2024 – Present
<i>End-to-End Platform Architecture, Modular Backend Services & Local AI Integration (Production Monorepo)</i>	
Architected Full-Stack Monorepo: Engineered scalable platform combining a responsive React client, modular Node.js/Express REST API (/api/v1), and a standalone C++ companion coaching engine. Engineered Local AI Companion Engine: Implemented persistent context memory (memory.json) tracking user health goals, workout consistency, and nutrition, achieving zero-cloud-cost personalized interaction. API & Community Architecture: Designed JWT authentication, user profile services, and interactive community post/comment endpoints with Vercel serverless integration and robust production error handling. Full Technical Ownership: Drove system design from database schemas and API contracts to UI implementation, performance profiling, and continuous deployment.	

FEATURED AI SECURITY & MULTI-AGENT CASE STUDY

AegisSwarm — Autonomous AI Red-Teaming & Security Framework	2025 – 2026
<i>Python, Multi-Agent Orchestration, Event-Driven Architecture, LLM Security (Research & Prototype)</i>	
Autonomous Multi-Agent Red-Teaming: Architected an event-driven framework to systematically simulate and evaluate adversarial security attacks against LLM agents and multi-agent systems. Advanced Planning & Attack Chains: Integrated Hierarchical Task Networks (HTN) and Tree-of-Thoughts (ToT) reasoning loops for coordinated multi-step attack chain exploration and evaluation. Threat Model Simulation: Implemented attack modules probing indirect prompt injection via tools, long-term memory poisoning, RAG corruption, and unauthorized tool invocations. Security Benchmarking: Evaluated agent defenses against standardized security benchmark suites (ASB, AgentDojo, HarmBench) to generate structured automated vulnerability reports.	

SELECTED AI, SYSTEMS & SOFTWARE PROJECTS

GrindHaus AI — Scientific RAG Pipeline & Custom PyTorch Transformer

2025 – 2026

Python, PyTorch, FAISS (C++ Vector DB), SentenceTransformers, REST API (Implemented & Validated)

- **Scientific RAG Ingestion Pipeline:** Engineered document extraction for sports science research papers: sliding-window chunking, all-MiniLM-L6-v2 dense embeddings, and native C++ FAISS vector indexing for grounded retrieval.
- **Custom Transformer Architecture:** Implemented a 20-layer PyTorch GPT-style model (hidden_size=1536, 12 heads, max_seq_len=2048) featuring RoPE positional embeddings, SwiGLU activations, RMSNorm, KV-cache inference, and bf16 training scripts.
- **Curated Dataset & Dynamic Router:** Generated and deduplicated an 800-sample domain instruction dataset with an intent-based router dynamically switching between Expert (scientific) and Companion (conversational) modes.

Real-Time Voice AI Streaming Pipeline

2025

Node.js, WebSockets, Bidirectional Audio Streaming, VAD, STT, TTS (Implemented & Tested)

- **Low-Latency Streaming Pipeline:** Engineered bidirectional streaming connecting browser audio to backend using WebSockets with ~20ms audio chunking for low-latency transmission.
- **Modular AI Processing Stages:** Integrated energy-based Voice Activity Detection (VAD), Speech-to-Text (STT), streaming LLM reasoning, and Text-to-Speech (TTS) with parallelized execution stages for fast response generation.

PHFIXT / Client Platform (Therapist Website)

2025

Next.js, Tailwind CSS, Vercel Cloud, Responsive Architecture (Production Deployed)

- **Production Web Application:** Engineered a full-stack responsive web platform with modular components, custom typography scale, sub-second load times, and cloud deployment on Vercel.
- **Accessible Layout & Performance:** Implemented mobile-first layout systems, micro-interactions, and optimized assets achieving high Lighthouse performance scores.

ExpenJar — Offline-First Android Expense Management App

2024

Java, Android SDK, SQLite Room Database, MVVM Architecture (Implemented)

- **Native Mobile Architecture:** Built an offline-first financial tracker using SQLite via Room Database for persistent transactional storage and category-wise spending analytics.
- **State & Data Integrity:** Designed reactive UI components with transaction rollback handling and structured monthly financial breakdown visualizations.

ENGINEERING METHODOLOGY & TECHNICAL EXECUTION

End-to-End Execution: 01 Understand requirements & constraints → 02 Architect data flow & interfaces → 03 Build core functionality → 04 Integrate APIs, DBs & AI services → 05 Validate, benchmark & test → 06 Package & Deploy → 07 Monitor & Iterate. Focused on pragmatic architectural decision-making, minimal technical debt, and rapid problem-solving across any layer of the software stack.

EDUCATION

Government College of Engineering, Chandrapur

Expected 2026

Bachelor of Technology (B.Tech) in Computer Science & Engineering

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Software Engineering, Object-Oriented Programming